



BUSINESS WHITEPAPER & PRESENTATION BRIEF

SOLARQUOTEPRO MASTER BRIEFING & USER GUIDE

Automating Solar Sales, Sizing, and Margin Protection in the
Nigerian Energy Ecosystem

1. Executive Summary

SolarQuotePro is a specialized, value-first transactional SaaS and Customer Relationship Management (CRM) platform built to address the unique challenges of the Nigerian solar energy market.

The Value Proposition

SolarQuotePro acts as a business-in-a-box for installers and an educational, transparent pricing agent for homeowners, helping solar contractors eliminate manual quoting, protect their profit margins from inflation, and close high-ticket deals in minutes rather than days.

By automating sizing, detailing generator displacement return on investment (ROI), and offering dynamic USD-to-Naira exchange rate buffers, the platform bridges the trust gap between installers and property owners.

2. The Problem Landscape

Operating a solar business or purchasing a solar system in Nigeria is plagued by structural, economic, and logistical friction:

A. The Fuel Subsidy Shock

Following the removal of petrol subsidies and the floating of the Naira, fuel prices soared to **₦1,200+ per liter**. Running a generator is no longer a standard operational expense; it is a financial emergency. Simultaneously, grid electricity prices (Band A tariffs) have risen dramatically, squeezing household and business budgets.

B. High Capital Requirements & The "Sticker Shock"

Transitioning to hybrid solar requires substantial upfront capital. When presented with quotes in the millions of Naira, property owners experience sticker shock. Because they do not understand how the system capacity is calculated or how it offsets fuel costs, they hesitate to buy, stalling the installer's sales pipeline.

C. Naira Volatility & FX Risk

Solar hardware (inverters, tier-1 panels, lithium-iron-phosphate batteries) is imported and priced in US Dollars (USD). However, installers must present their quotes in Nigerian Naira (NGN). Since the Naira fluctuates daily, a quote prepared on Monday can become unprofitable by Friday. Installers are caught between losing deals by overpricing or losing money by underpricing.

D. Administrative and Design Inefficiencies

Independent contractors and local engineering firms waste hours manually designing load profiles in Excel, looking up prices from various suppliers, and designing static PDF proposals. These static files are hard to edit, lack interactive elements, and fail to project a modern, professional corporate image.

3. The SolarQuotePro Solution

SolarQuotePro provides a "**Value-First**" automated platform that removes friction from the solar sales cycle:

- **Progressive Disclosure:** Instead of forcing homeowners to sign up or fill out long forms, it provides a zero-setup, interactive **Quick Sizer** that gives instant answers.
- **Diesel/Petrol Displacement Math:** It translates raw solar panel counts and inverter sizes into tangible money saved on generator fuel, converting "expenditure" into an "investment."
- **Dynamic FX Buffers:** It allows installers to set a custom exchange rate (official vs. parallel-market parallel rate) with safety limit protections, ensuring they never quote below cost.
- **Instant Payments via Paystack:** It embeds secure payment buttons directly inside interactive online proposals, enabling clients to pay down-payments or book site surveys on the spot.

4. Deep-Dive Feature Catalog

Below is a summary of the core modules integrated into the platform:

FEATURE	DESCRIPTION	BUSINESS IMPACT
 Simple Mode Sizer	A client-facing tool that matches typical Nigerian appliances with backup times.	Acts as a high-conversion lead magnet, driving organic traffic.
 Generator ROI Engine	Calculates monthly savings and payback periods by contrasting fuel expenses against solar amortization.	Overcomes client skepticism by proving exactly when the solar system pays for itself.
 Inflation-Guard FX	Allows installers to adjust Naira quotes instantly using custom currency rates with safety boundaries.	Eliminates FX losses, protecting profit margins during exchange rate spikes.
 Web Proposals	Generates secure web links where clients can toggle loads live and see changes in pricing and backup times.	Enhances professional credibility and dramatically shortens the negotiation loop.
 Paystack Checkout	Connects local bank transfers, cards, and USSD payments directly into the proposal pages.	Streamlines cash flow, allowing installers to collect site-survey fees instantly.
 Pro Workspace CRM	A pipeline dashboard designed to track proposals from draft to final installation.	Keeps sales organized, ensuring no hot lead gets forgotten.
 WhatsApp Sharing	Formats proposal links and summaries into clean messages optimized for WhatsApp.	Fits the primary business communication channel used by Nigerian clients.

5. Step-by-Step System Walkthrough

Here is exactly how different users navigate and interact with the main features of the platform:

Scenario A: The Property Owner Sizing their Building

1. Open the homepage (/) and navigate to the 🏠 **Sizer Estimator** section.
2. Select your building type (e.g., Residential Home or SME Office).
3. Tap on the icons of appliances you want to run on solar backup (e.g., 1 refrigerator, 4 LED light bulbs, 1 smart TV, 1 inverter air conditioner).
4. Select the estimated hours of runtime required at night (e.g., 6 hours) and click **Calculate Sizing**.
5. The app displays your recommended setup (e.g., 5kVA Inverter + 6 Panels + 1 Lithium Battery), estimated monthly savings (e.g., ₦180,000 saved), and the payback period.

Scenario B: The Solar Installer Creating an Inflation-Proof Quote

1. Go to /workspace and log in (e.g., installer@test.local).
2. On your workspace dashboard, click **Create New Proposal**.
3. Set your **FX Pricing Guard** by entering your desired exchange rate (e.g., ₦1,650/USD) in the FX field to hedge against currency spikes.
4. Add items to your Bill of Materials (BOM) from the equipment catalog. All prices update dynamically in Naira.
5. Set down-payment details and click **Publish Proposal**. Tap **Share via WhatsApp** to send the secure proposal link to the customer.

Scenario C: The Client Reviewing and Paying

1. The client opens the secure proposal link on their phone.
2. They use the **Interactive Load Toggle** to see how turning off specific appliances increases battery backup time and changes the total price.

3. To secure the installation, they click **Book & Pay Site Survey via Paystack** or pay the down-payment fee.

4. Paystack handles the secure card or bank transfer, and the installer's Kanban CRM updates the status automatically.

Scenario D: Platform Administrator Managing Currency Settings

1. Go to /admin and log in with your administrative account (e.g., admin@test.local).

2. Open the **FX Management Panel** (/admin/fx-rates).

3. Enter the custom parallel market base rate. The safety check automatically blocks rate inputs that are out of bounds (to avoid typing mistakes).

6. Business & Monetization Model

SolarQuotePro implements a tiered B2B SaaS model integrated with local payments (Paystack):

- **Free Tier (₦0/month):** Basic lead sizing and watermarked PDF quote exports.
- **Starter Tier (₦15,000/month):** Up to 5 unwatermarked quotes, custom local component pricing, and FX settings.
- **Pro Tier (₦35,000/month):** Unlimited quotes, custom branding templates, interactive web proposal checkouts, and installment plans.
- **Business Tier (₦75,000/month):** Full CRM pipeline tracking, WhatsApp sharing integrations, and multi-seat team collaboration.
- **Transactional Commissions:** A 1.5% transaction processing fee on site-survey bookings or deposits routed through the platform.
- **Supplier Catalogs:** Referral commission (2–5%) on hardware procurement orders routed directly to major wholesale importers.